Review retrospective:
the generic-k J-function, and a docstring that out-claimed its own hypotheses

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2026-06-05

Abstract

A soundness review of `Laplace/OneD/JFunctionMonomial.lean`: the generic-k J-function asymptotic $t^{1/(2k)} J_k(t) \to \pi(0) \cdot \frac{1}{k} \cdot ((2k)!)^{1/(2k)}\, \Gamma(1/(2k))$ that folds the quartic and sextic special cases into one parametric theorem. Result: **a clean fidelity pass with five cleanups** — a `show`→`change`, two redundant imports, and two docstring inaccuracies (a hypothesis the header over-stated, and a lemma-usage miscount).

1 Setting

This is the abstraction the sextic J-function specialises (and the quartic one should). Its design goal was the *minimal* hypothesis class — the prior need only be `AEMeasurable`, `ContinuousAt 0`, and globally bounded — and the review's most interesting finding was that the module docstring did not reflect that achievement.

2 What was reviewed

Seven declarations: the universal inequality $x^2 \le 1 + x^{2k}$, monomial-Gibbs integrability, the substitution lemma, the dominator integrability, the centred DCT theorem, the closed-form partition function, and the asymptotic corollary.

3 Fidelity / soundness findings

Sound; no mismatch. The partition closed form $(1/k)((2k)!/t)^{1/(2k)}\Gamma(1/(2k))$ is correct (the $2 \cdot (1/p_R) = 2/(2k) = 1/k$ step), the substitution ($a^{2k} = t$, $|a^{-1}| = 1/a$ cancelling the outer a) is right, and the limit constant matches. The load-bearing check was the DCT *measurability* step: with only `AEMeasurable` on the prior, the file gets `AEStronglyMeasurable` of $u \mapsto \pi(u/t^{\alpha})$ by composing with the quasi-measure-preserving scaling $u \mapsto (t^{\alpha})^{-1}u$ — so *no* global continuity is needed, and only `ContinuousAt 0` is used later for the pointwise limit. GPT confirmed this works as claimed; it is exactly why the sextic specialisation could inherit the weak hypotheses.

4 Simplifications applied

Five.

- A `show`→`change` on the reshaping goal in the measurability branch (cleared a standing `linter.style.show` warning — the same one flagged from the sextic review).
- Two redundant imports: `Laplace.Gibbs` (the file references no `partitionFunction`/`gibbsExpectation`/`gibbsCov` — `kth_partition` is stated as a raw integral) and `Gamma.Basic` (`Real.Gamma` arrives transitively via `Integral.Gamma`).
- Two docstring accuracy fixes. The module header advertised a "continuous, globally bounded prior", but the theorems take the weaker `AEMeasurable` + `ContinuousAt 0` + bound — the *whole point* of the refactor that produced this file. (The per-theorem docstring was already precise; only the module header lagged.) And the strategy prose claimed the master Gamma-integral lemma was "used twice", when in fact it is used once (the partition) and integrability is a separate `integrable_exp_neg_mul_sq` argument.

Foundation file (two consumers); arbitrated by a **full project build**, green at 8306 jobs with both consumers rebuilt, three public signatures byte-identical, file now warning-free; GPT affirmed all five.

5 Disagreements and open questions

None.

6 Lessons

- **A module header can lag the very refactor that file embodies.** This file exists to provide the minimal-hypothesis J-function, yet its top docstring still described the old "continuous" hypothesis — while the per-theorem docstrings, written later, were accurate. When a file's purpose is a hypothesis weakening, the summary at the top is a prime spot for stale over-statement; check it against the actual signatures.
- **"Used twice" is a checkable claim.** Prose that counts how often a lemma is invoked, or names which lemma does the work, is verifiable by reading the proof — and here it was wrong (the master rpow lemma once, Gaussian comparison separately). Strategy prose that names mechanisms should be audited like a theorem statement, not skimmed.

7 Follow-ups

- **Forward tide (carried from the sextic review):** migrate `JFunctionQuartic` to specialise this generic file, weakening its hypothesis from global `Continuous` to `AEMeasurable` + `ContinuousAt 0`.
- **Next review:** `JFunctionAsymptoticEquivalence.lean` (the remaining J-function file, and this file's other consumer).